MODULE 02

Contracting in Construction

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Contract

- Contract may be defined as an agreement which is enforceable by law.
- It is a written undertaking for execution of work or supply of materials or performance of any service.
- In simple terms, A contract is a mutual promise that the law will support. It ensures that if one party doesn't keep their word, the other party can seek justice through legal channels (like courts or arbitration).
- Example: If a construction contractor agrees to build a house for ₹50 lakhs and the client agrees to pay that amount upon completion, this forms a contract.

Purpose of Contract

- Clarity of Terms: To clearly define the rights, duties, and expectations of all parties involved.
- Legal Protection: To provide a legal remedy in case any party fails to fulfill their obligations.
- Risk Management: To minimize disputes and manage business or project risks effectively.
- Enforceability: To ensure that commitments made by parties are enforceable in a court of law.
- Documentation: To serve as written evidence of the agreement in case of future disagreements.

Sanctity of Contract

- Sanctity of contract means that when two or more people make a legal agreement, they must keep their promises. The law says that a valid contract should be respected and followed by everyone who signs it.

Importance:

- It helps people trust each other in business and personal deals.
- It makes sure that everyone does what they agreed to do.
- It helps the economy work smoothly, because people know their deals will be honored.
- It gives a legal way to solve problems if someone breaks the agreement.

Sanctity of Contract

Exceptions of Sanctity of Contract:

Though contracts are sacred, certain exceptions allow for modification or termination:

- Public Interest or Welfare(e.g., government overriding a contract for national security).
- Unconscionable Terms (e.g., one-sided or exploitative conditions).
- Force Majeure / Frustration(e.g., natural disasters making performance impossible).
- Contracts against law or morality.

Example: Even if the market price of materials rises later, the contractor cannot unilaterally demand a higher price, unless the contract has a clause for price escalation.

Types of Construction Contracts

Item Rate Contract

- Also called a schedule contract, in this contract, the contractor undertakes the execution of work on an item rate basis.
- The payment to the contractor is made on the basis of detailed measurements of different items of work actually donc by him.
- It is widely used in public works and government projects, where the scope of work is defined, but the quantities may vary during execution.
- BOQ is prepared by the client/consultant listing all items of work and contractor quotes rate for each item.

Item Rate Contract: (Merits)

- **Fair and Transparent Payment:** Contractor is paid based on actual quantity of work executed, ensuring fair compensation.
- **Detailed Cost Breakdown:** Each item is individually priced, allowing clear understanding and control of costs.
- **Flexibility in Quantity Variations:** Useful in projects where quantities are not fixed and may change during execution.
- **Competitive Tendering:** Promotes healthy competition as contractors bid item-wise, leading to cost-effective selection.
- **Easy to Compare Bids:** Client can compare item rates across bidders and evaluate suitability.

Item Rate Contract: (Demerits)

- **Requires Continuous Measurement:** Frequent site measurement and verification increase administrative burden.
- **Delayed Payments:** Time-consuming measurement and approval process can delay contractor billing.
- **Risk of Unbalanced Bids:** Contractors may quote very low rates for fixed items and high for items expected to vary.
- **High Supervision Needs:** Close monitoring is essential to check quality and correct quantity reporting.
- **Scope for Disputes:** Disagreements may arise over measurement, rates for extra items, or quantity changes.

Lump Sum Contract

- In this type of contract, the contractor offers to do the whole work as shown in drawings and described by specifications, for a total stipulated sum of money.
- There are no individual rates quoted, thus it becomes difficult to make adjustments in the contract value if any changes are to be made in the work later on.
- The schedule of different items of work is not provided, and the contractor has to complete the work as per drawings and specifications for the agreed lump sum amount.

Lump Sum Contract

- Deposit of 10% security money and other conditions of the contract are included in the contract agreement.
- Upon the completion of work, a fixed lump sum amount is paid to the contractor.
- Detailed measurements of different items are required but the whole work is compared and checked with drawings and specifications before releasing the payment.
- In large projects, part payments are made to the contractor at different stages of work on mutually agreed terms. In case the contractor stops the work in between he is not entitled for any further payment.

Lumpsum Contract: (Merits)

- The owner can decide whether to start or shelve the project knowing the total lump sum price quoted by different contractors.
- The risk of cost overruns due to delays or increased material costs lies mostly with the contractor, not the client.
- Fewer progress measurements and change order negotiations are required, making project management simpler.
- Fixed price may motivate the contractor to complete the work ahead of time to maximize profits.

Lumpsum Contract: (Demerits)

- Before the contract is awarded, the project has to be studied thoroughly and the complete contract document has to be prepared in advance.
- Any changes in design or scope after signing the contract require renegotiation and may lead to cost escalation.
- Disagreements may arise if the drawings/specifications are not clear, leading to conflicts about what is included in the contract.
- To maximize profit, the contractor might use cheaper materials or cut corners if not properly supervised.

Cost Plus Fixed Fee Contract

- A Cost Plus Fixed Fee Contract is an agreement in which the contractor is paid for all incurred expenses plus a pre-agreed fixed amount as profit.
- In order to negotiate such a type of contract, it is essential that the scope and some general details of the work are defined.
- The contractor in this type of contract is selected on the basis of merit rather than the fee alone.
- Once this fee is fixed, the contractor cannot increase the cost of work.
- This contract is also suitable for important structures where the cost of construction is immaterial.

Cost Plus Fixed Fee Contract: (Merits)

- In this type of contract, actual cost is to be borne by the owner. Therefore, the contractor performs the work in the best interest of the owner resulting in good quality work.
- The work can be taken in hand even before the detailed drawings and specifications are finalized.
- Changes in design and method of construction if needed can be easily carried out without disputes.
- The work can be executed speedily.

Cost Plus Fixed Fee Contract: (Demerits)

- This form of contract cannot be adopted normally in case of public bodies and Government departments.
- The final cost of the work is no known in advance and this may subject the owner to financial difficulties.
- Tracking and approving costs can be resource-intensive which may lead to higher administrative cost.

Cost Plus Percentage of Cost Contract

- In this type of contract, instead of awarding the work on lumsum or item rate basis, it is given on certain percentage over the actual cost of construction.
- The actual cost of construction is reported by the contractor and paid to him by the owner together with a certain percentage as agreed earlier.
- The contractor agrees to do the work in accordance with the drawings, specifications, other conditions of contract.
- In this type of contract proper control has been exercised by the owner in purchase of materials and in arranging labour.

Cost Plus %of Cost Contract: (Merits)

- The project can begin without waiting for detailed designs and estimates.
- Suitable for projects with uncertain or changing requirements.
- All actual costs are shared with the client, ensuring clarity.
- Since cost is reimbursed, the contractor may focus more on quality than cutting corners.
- Contractor may try new methods or materials without fearing upfront loss.

Cost Plus %of Cost Contract: (Demerits)

- Final cost is unknown until completion, which may affect budgeting.
- Contractor has less incentive to reduce expenses since higher cost means higher fee.
- Owner must review and approve ongoing expenditures.
- Possibility of inflated billing or inefficient resource use.
- Differences in interpretation of allowable costs can lead to disputes.

Turnkey Contract

- A Turnkey Contract is an agreement where the contractor assumes full responsibility for the entire lifecycle of a project including design, procurement, construction, testing, and commissioning.
- The project is delivered in a ready-to-operate condition, meaning the client can immediately start using the facility with minimal additional work.
- The term “Turnkey” implies that the client only needs to "turn the key" to begin using the facility.
- IT parks developed on turnkey basis by L&T or Shapoorji Pallonji are some of the example of the turnkey project.

Turnkey Contract: (Merits)

- The contractor assumes full responsibility, reducing the client's risk.
- Minimal client involvement is needed, saving time and resources.
- Fixed price offers cost certainty and better budget control.
- Faster project delivery due to streamlined coordination under one contractor.

Turnkey Contract: (Demerits)

- Limited flexibility for the client to make changes during execution.
- Less control over design and material choices may affect quality.
- Higher upfront cost due to contractor pricing for risk coverage.

Contract Documents

- **Title Page:** It contains the name of the work, name of the owner, name of the contractor, contract agreement number, contents etc.
- **Contents Page:** It contains the contents of the agreement with page references
- **Notice Inviting Tender(NIT):** It contains a brief description of work estimated cost of work date and time for receiving the tender, amount of earnest money (EMD), security deposit(SD), time of completion etc.
- **Tender form:** It comprises of bill of quantities, contractor‘s rate, total cost of work, time for completion, security money to be deposited and penalty process etc.

Contract Documents

- **Schedule of issue of materials:** It contains the list of materials to be issued by the department or owner to the contractor with rates or place of issue.
- **Drawings:** These comprise a complete set of fully dimensioned drawings including plans, elevations, sections, detailed drawings and site plan.
- **Specifications:** Following specifications are normally included in the contract document

a. General specification- These specify the class and type of work, quality of materials etc., in general for the work as a whole.

b. In detailed specification – These views detailed description of each item of work including material and method to be used along with quality of workmanship required

Contract Documents

- **Conditions of Contract:**

a. Rates of each item of work inclusive of materials, labour , transport, plant and equipment and other arrangements required for the completion of work.
b. Amount and form of earnest and security money to be deposited
c. Manner of payment to contractor including running payments, final payments, refund of security money
d. Time of completion of work
e. Proportionate progress to be achieved
f. Penalty for poor quality and unsatisfactory work, and for delay in completion
g. Extension of time for completion of work
h. Engaging other agency at contractors cost and risk
i. Termination of contract
j. Changes in design and drawings etc. and valuation of variations
k. Measurement of the work
l. Arbitration for settlement of disputes
m. In addition to the above performance and payments bonds are also sometimes considered as part of the contract document.

What is FIDIC?

- FIDIC is an international standards organization best known for its suite of contracts used for construction and engineering projects globally.
- Its standard contract forms are widely used for both domestic and international infrastructure projects, particularly where parties come from different legal and cultural backgrounds.

FIDIC Contract Types

- **Red Book** – Construction (Building & Engineering Works) – contractor builds to the employer's design.
- Yellow Book – Plant and Design-Build – contractor is responsible for design and execution.
- Silver Book – EPC/Turnkey Projects – contractor bears more risk, including design and performance.
- Green Book – Short Form of Contract – used for small-scale projects.
- **Gold Book** – Design, Build, and Operate – includes long-term operation/maintenance.
- White Book – Client/Consultant Model Services Agreement.

Contract Management

- Contract Management is the process of managing contract creation, execution, and analysis to maximize operational and financial performance while minimizing risk.
- It is especially critical in fields like construction, engineering, procurement, IT services, and legal sectors.
- Good contract management helps all parties clearly understand their responsibilities, timelines, and payments, which leads to fewer disputes and smoother project completion.



Key Components of Contract Management

- **Contract Creation** – Drafting contracts based on project and legal requirements.
- **Negotiation** – Finalizing terms, conditions, and responsibilities of all parties.
- **Execution** – Signing and implementing the contract as agreed.
- **Performance Monitoring** – Tracking progress, deliverables, and compliance.
- **Change Management** – Managing changes like variations and time extensions.
- **Dispute Resolution** – Handling claims, conflicts, and legal issues.
- **Contract Closure** – Completing final payments, documentation, and reviews.

THANK YOU